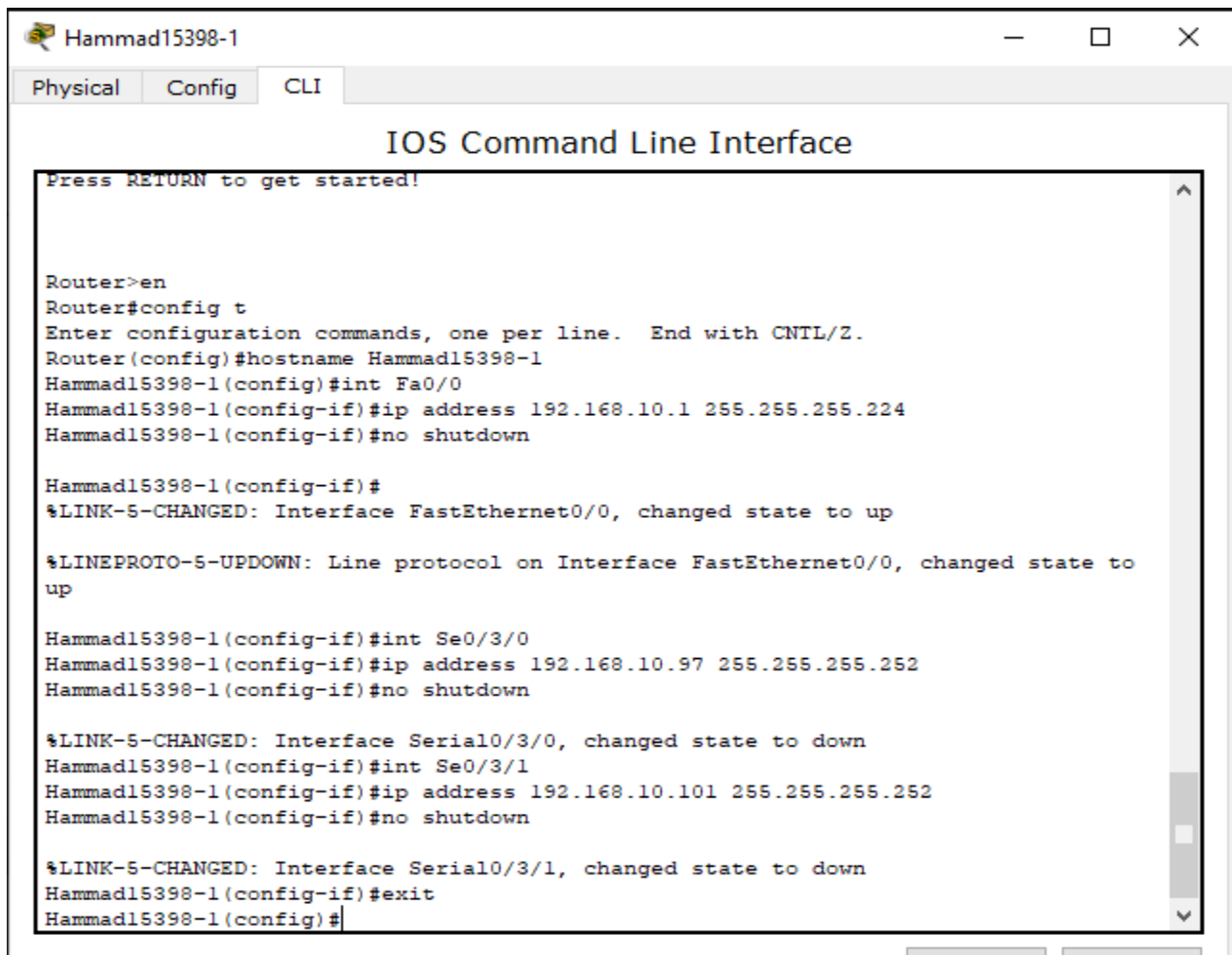




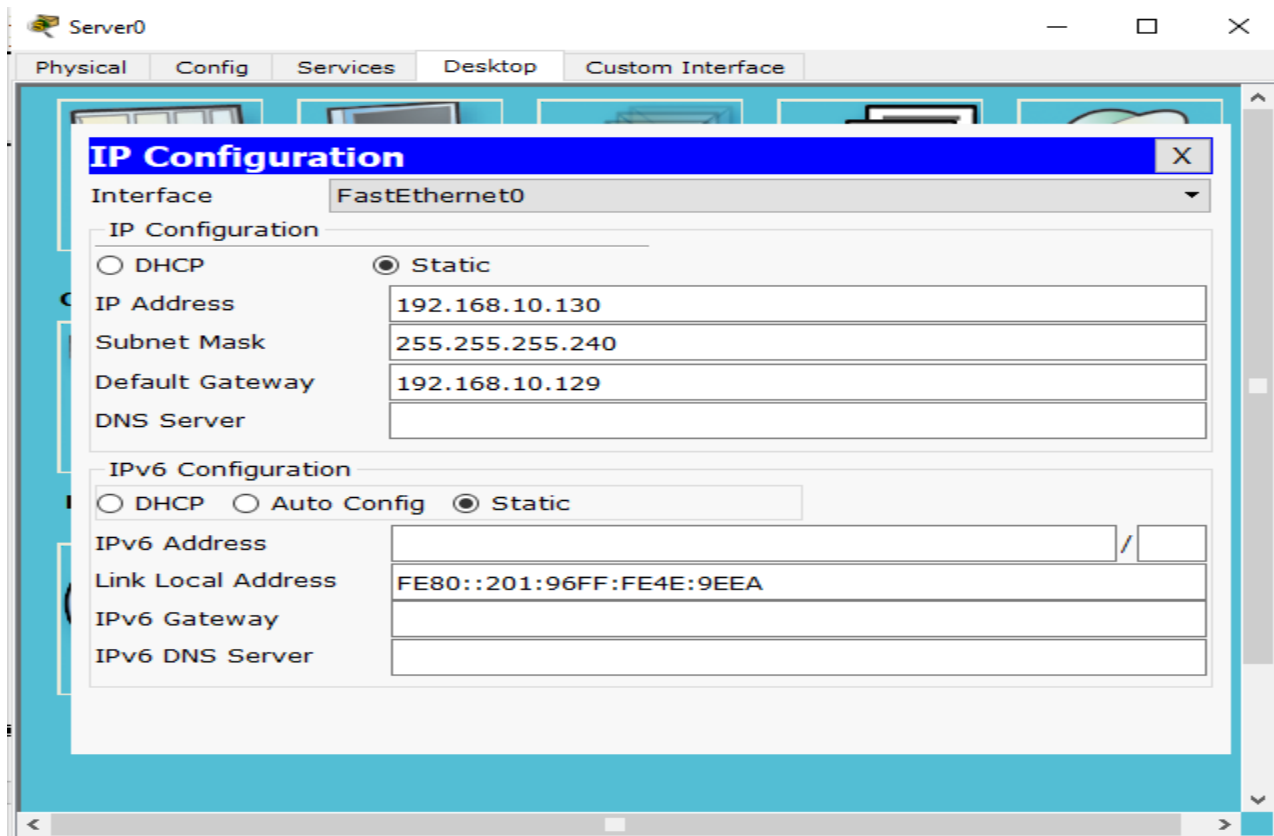
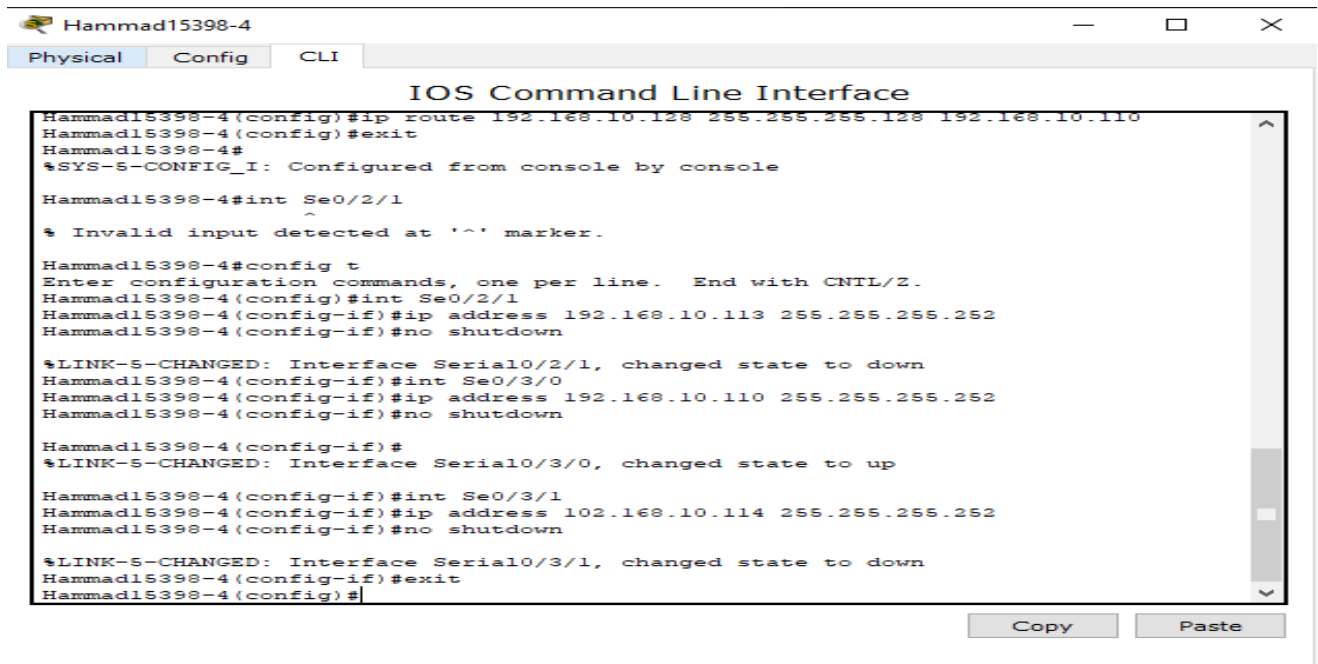
 Hammad15398-2

Physical Config CLI

```
239K bytes of non-volatile configuration memory.
62720K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE SOF
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 06:21 by pt_rel_team
```

Press RETURN to get started!

```
Hammad15398-2>en
Hammad15398-2#config t
Enter configuration commands, one per line. End with CNTL/Z.
Hammad15398-2(config)#int Fa0/0
Hammad15398-2(config-if)#ip address 192.168.10.33 255.255.255.224
Hammad15398-2(config-if)#no shutdown
Hammad15398-2(config-if)#
%LINK-S-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Hammad15398-2(config-if)#int Se0/3/0
Hammad15398-2(config-if)#ip address 192.168.10.98 255.255.255.252
Hammad15398-2(config-if)#no shutdown
Hammad15398-2(config-if)#
%LINK-S-CHANGED: Interface Serial0/3/0, changed state to up
Hammad15398-2(config-if)#int Se0/3/1
Hammad15398-2(config-if)#ip address 192.168.10.102 255.255.255.252
Hammad15398-2(config-if)#no shutdown
Hammad15398-2(config-if)#
%LINK-S-CHANGED: Interface Serial0/3/1, changed state to up
Hammad15398-2(config-if)#int Se0/2/0
Hammad15398-2(config-if)#ip address 192.
%LINEPROTO-S-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
^
% Invalid input detected at '^' marker.
Hammad15398-2(config-if)#int Se0/2/0
Hammad15398-2(config-if)#ip address 192.168.10.105 255.255.255.252
Hammad15398-2(config-if)#no shutdown
%LINK-S-CHANGED: Interface Serial0/2/0, changed state to down
Hammad15398-2(config-if)#exit
Hammad15398-2(config)#
```



Server0

Physical Config Services Desktop Custom Interface

**SERVICES**

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP

**DHCP**

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: LAN5

Default Gateway: 192.168.10.161

DNS Server: 0.0.0.0

Start IP Address : 192 168 10 162

Subnet Mask: 255 255 255 240

Maximum number of Users : 14

TFTP Server: 0.0.0.0

Add Save Remove

| Pool Name | Default Gateway | DNS Server | Start IP Address | Subnet Mask | Max User | Time  |
|-----------|-----------------|------------|------------------|-------------|----------|-------|
| LAN7      | 192.168.10.193  | 0.0.0.0    | 192.168.10....   | 255.255.... | 14       | 0.0.0 |
| LAN6      | 192.168.10.177  | 0.0.0.0    | 192.168.10....   | 255.255.... | 14       | 0.0.0 |
| LAN5      | 192.168.10.161  | 0.0.0.0    | 192.168.10....   | 255.255.... | 14       | 0.0.0 |
| LAN4      | 192.168.10.145  | 0.0.0.0    | 192.168.10....   | 255.255.... | 14       | 0.0.0 |

Hamad15398-1

Physical Config CLI

### IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Hamad15398-1(config-if)#int Se0/3/0
Hamad15398-1(config-if)#ip address 192.168.10.97 255.255.255.252
Hamad15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Hamad15398-1(config-if)#int Se0/3/1
Hamad15398-1(config-if)#ip address 192.168.10.101 255.255.255.252
Hamad15398-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Hamad15398-1(config-if)#exit
Hamad15398-1(config)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Hamad15398-1(config)#ip route 192.168.10.32 255.255.255.224 192.168.10.98
Hamad15398-1(config)#ip route 192.168.10.64 255.255.255.224 192.168.10.98
Hamad15398-1(config)#ip route 192.168.10.32 255.255.255.224 192.168.10.102 10
Hamad15398-1(config)#exit
Hamad15398-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

PC6

Physical Config Desktop Custom Interface

### IP Configuration

IP Configuration

☒ DHCP ☐ Static

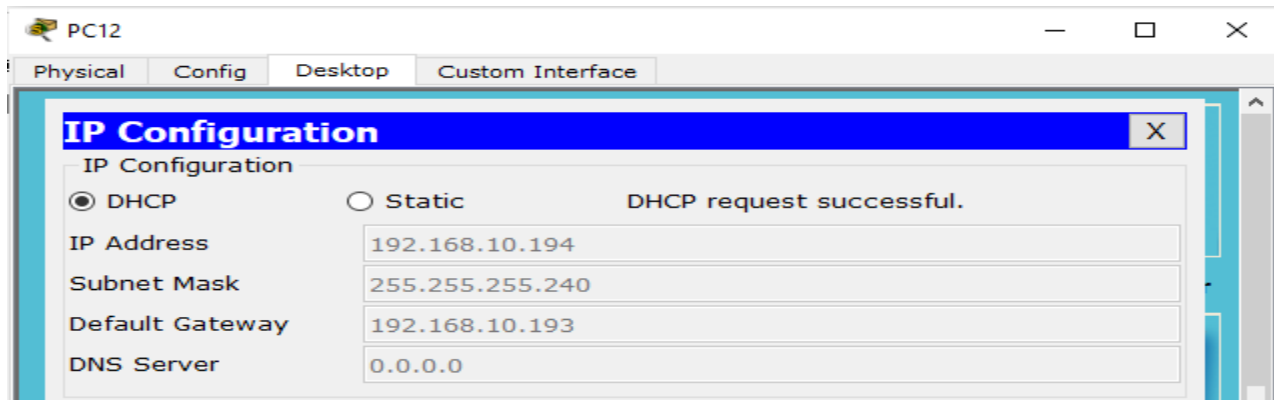
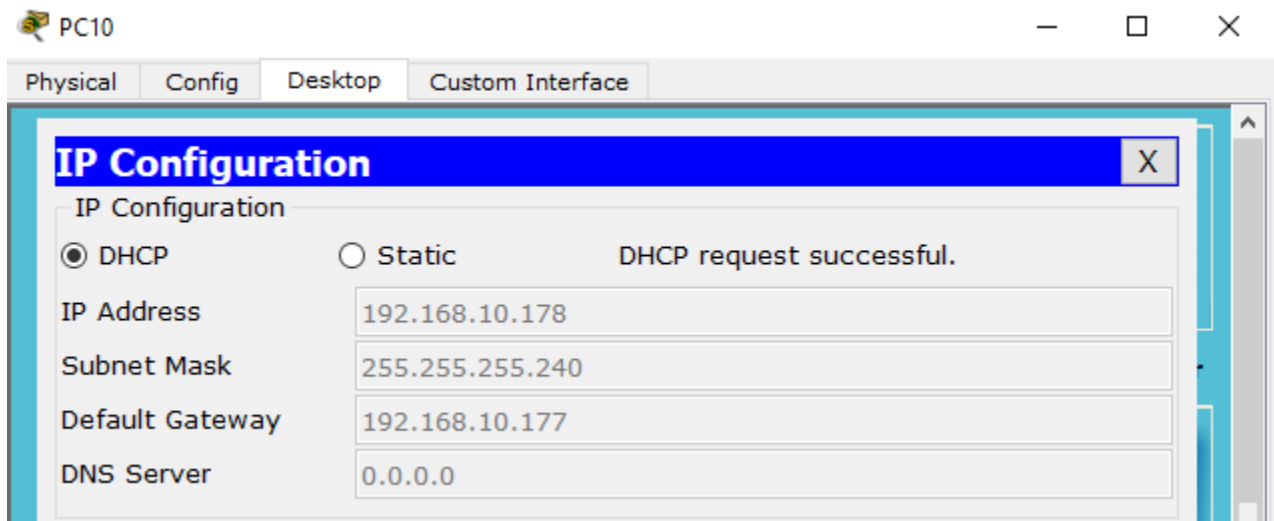
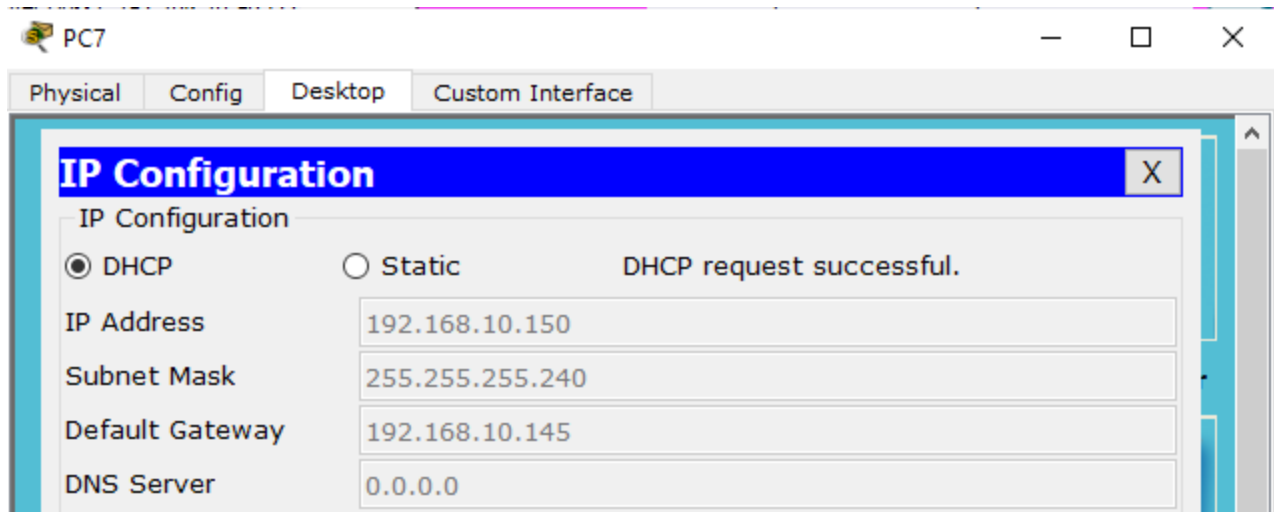
IP Address 192.168.10.146

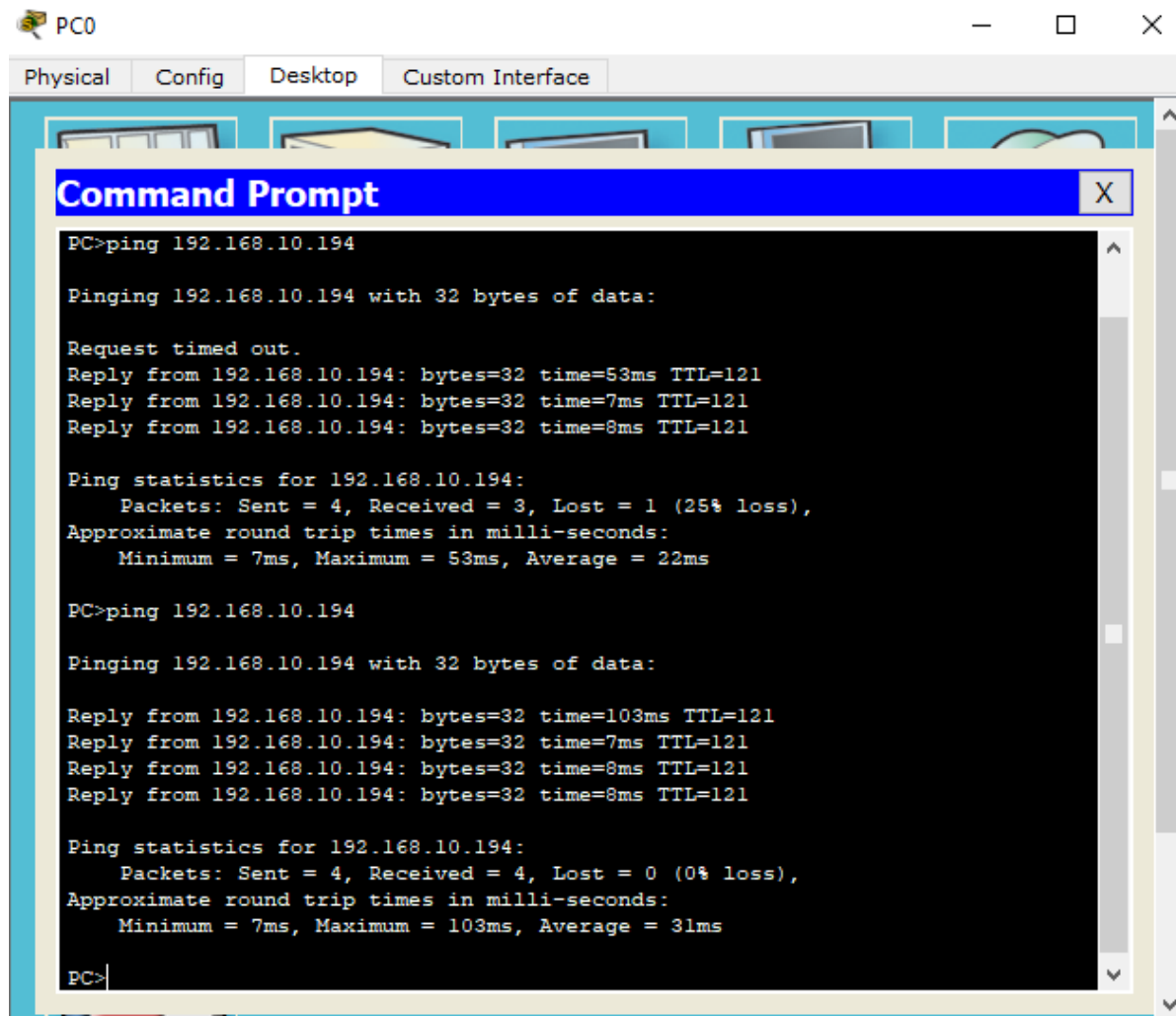
Subnet Mask 255.255.255.240

Default Gateway 192.168.10.145

DNS Server 0.0.0.0

IPv6 Configuration





The screenshot shows a Packet Tracer PC0 desktop environment. A 'Command Prompt' window is open, displaying the results of two ping commands to the IP address 192.168.10.194. The first ping attempt shows a 'Request timed out' for the first packet, while the subsequent three packets are received successfully. The second ping attempt shows all four packets being received successfully. The window title bar includes 'Physical', 'Config', 'Desktop', and 'Custom Interface' tabs.

```
PC>ping 192.168.10.194

Pinging 192.168.10.194 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.194: bytes=32 time=53ms TTL=121
Reply from 192.168.10.194: bytes=32 time=7ms TTL=121
Reply from 192.168.10.194: bytes=32 time=8ms TTL=121

Ping statistics for 192.168.10.194:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 53ms, Average = 22ms

PC>ping 192.168.10.194

Pinging 192.168.10.194 with 32 bytes of data:

Reply from 192.168.10.194: bytes=32 time=103ms TTL=121
Reply from 192.168.10.194: bytes=32 time=7ms TTL=121
Reply from 192.168.10.194: bytes=32 time=8ms TTL=121
Reply from 192.168.10.194: bytes=32 time=8ms TTL=121

Ping statistics for 192.168.10.194:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 103ms, Average = 31ms

PC>
```

## Scenario Selection & Description:

**Scenario Chosen:** Multi-Branch Corporate Network with Centralized Server Infrastructure.

**Headquarters / Main Branch:** Yahan par ahem servers (jaise DHCP Server) hosted hain jo poore network ko dynamically IP addresses faraham karte hain.

**Sub-Branche:** Network ko do hisson mein taqseem kiya gaya hai:

☐ **Static Routing (FLSM) Zone:** Jahan fixed subnetting ke zariye branches aapas mein connected hain aur security ya traffic control ke liye static routes manually configure kiye gaye hain.

□ **Dynamic Routing (RIPv2 VLSM) Zone:** Jahan variable length subnet masks (VLSM) ke zariye IPs ki bachat ki gayi hai aur dynamic routing protocol (RIPv2) ka istemaal kar ke networks automatic update hote hain.

### Network Topology & Addressing Scheme

Topology ko do major routing concepts (FLSM aur VLSM) ko demonstrate karne ke liye design kiya gaya hai.

#### Subnetting Details (IP Plan):

| Zone / Network    | Network Address    | Subnet Mask     | Description / Connected Devices                |
|-------------------|--------------------|-----------------|------------------------------------------------|
| LAN 1             | 192.168.10.0 /27   | 255.255.255.224 | Switch0 (PC0, PC1)                             |
| LAN 2             | 192.168.10.32 /27  | 255.255.255.224 | Switch1 (PC2, PC3)                             |
| LAN 3             | 192.168.10.64 /27  | 255.255.255.224 | Switch2 (PC4, PC5)                             |
| WAN (FLSM Links)  | 192.168.10.96 /27  | 255.255.255.224 | Router to Router serial links (e.g., .97, .98) |
| Server Network    | 192.168.10.128 /28 | 255.255.255.240 | Switch3 (Central DHCP Server0)                 |
| LAN 4             | 192.168.10.144 /28 | 255.255.255.240 | Switch4 (PC6, PC7)                             |
| LAN 5             | 192.168.10.160 /28 | 255.255.255.240 | Switch5 (PC8, PC9)                             |
| LAN 6             | 192.168.10.176 /28 | 255.255.255.240 | Switch6 (PC10, PC11)                           |
| LAN 7             | 192.168.10.192 /28 | 255.255.255.240 | Switch7 (PC12, PC13)                           |
| WAN (RIPv2 Links) | 192.168.10.224 /30 | 255.255.255.252 | Dynamic Zone WAN Router Point-to-Point links   |

## Protocol Implementation & Device Configuration

Screenshots ke mutabiq darj-zail major protocols aur configurations implement ki gayi hain:

### A. Centralized DHCP Service

- **Server0** ko statically IP 192.168.10.130 assign kiya gaya hai jiska default gateway 192.168.10.129 hai.
- Server par **DHCP Service "On"** hai aur multiple pools (LAN4, LAN5, LAN6, LAN7) banaye gaye hain taake end devices ko automatic IPs mil sakein (Jaise PC6 ko 192.168.10.146 aur PC12 ko 192.168.10.144 mila).

